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Modeling exercise 2

Production of three products

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Production of three products

A company produces three different products [A, B, C] in six different factories. The unitary production cost of each product and the capacities of each factory are given in the table below:

	F1	F2	F3	F4	F5	F6
A	25	30	26	34	32	30
B	30	32	34	35	38	40
C	40	46	42	37	40	50
Capacity	550	700	1100	350	400	450

Each quantity of these products can be sold—a continuous quantity, but the marketing department has already sold 700 units of product A, 500 units of product B, and 600 units of product C via contracts. The selling price of each of the products [A, B, C] is 60, 82.5, and 108€, respectively.

Propose a model that yields a production plan which maximizes profits

Let's identify the objective function

A company produces three different products [A, B, C] in six different factories. The unitary production cost of each product and the capacities of each factory are given in the table below:

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Let's identify the decision variables

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Let's identify the constraints

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Let's identify the sets

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Let's identify the parameters

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Time to formulate

SETS

p : products $\{A, B, C\}$

f : factories $\{F1, F2, F3, F4, F5, F6\}$

PARAMETERS

Q_f : capacity of factory f [unit]

$SOLD_p$: contract of product p [unit]

SP_p : selling price of product p [€]

$C_{p,f}$: unitary cost of product p in factory f [€/unit]

VARIABLES

$q_{p,f}$: quantity of product p in factory f [unit]

OBJECTIVE FUNCTION

Maximize profits [€]

$$\max \left[\sum_{p,f} SP_p \cdot q_{p,f} - \sum_{p,f} C_{p,f} \cdot q_{p,f} \right]$$

Diagram: An orange box highlights the first term, and a blue box highlights the second term. An orange arrow points from the orange box to a factory icon and an orange square labeled 'max'. A blue arrow points from the blue box to a cow icon and a blue square labeled 'min'.

CONSTRAINTS

Maximum capacity of factory f [unit]



$$\sum_p q_{p,f} \leq Q_f \quad \forall f$$

Minimum sold product p via contract [unit]



$$\sum_f q_{p,f} \geq SOLD_p \quad \forall p$$